higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS TECHNOLOGY
NQF LEVEL 3

(6050353)

26 February 2019 (X-Paper)
09:00–12:00

Nonprogrammable calculators may be used.

This question paper consists of 5 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 Fluids mix with each other in heat exchangers.

1.2 The main advantage of an air cooler/heater is that it requires cooling water from a cooling tower.

1.3 A helical shape is a spiral shape.

1.4 Colloids are particles that settle out due to gravity.

1.5 Osmosis is the process by which substances move across a membrane from an area of high concentration to one of low concentration.

(5 × 1) **[5]**

QUESTION 2

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (2.1–2.5) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1	Material used to cover fins in air coolers/heaters	A	convection
2.2	Type of dry-cooling tower	B	hybrid cooling towers
2.3	Method of drying where hot air is blown over a wet material	C	electrolytic cell
2.4	Most adsorbents with very fine pores	D	conduction
2.5	Electrochemical cell that produces electric current from the decomposition of substances	E	zinc
		F	desiccants
		G	voltaic cell
		H	aluminium
		I	molecular sieves
		J	indirect dry-cooling tower

(5 × 1) **[5]**

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QUESTION 3

3.1 Define each of the following terms:

3.1.1 Viscosity (2)

3.1.2 Safety data sheet (3)

3.1.3 Sampling (2)

3.1.4 Density (2)

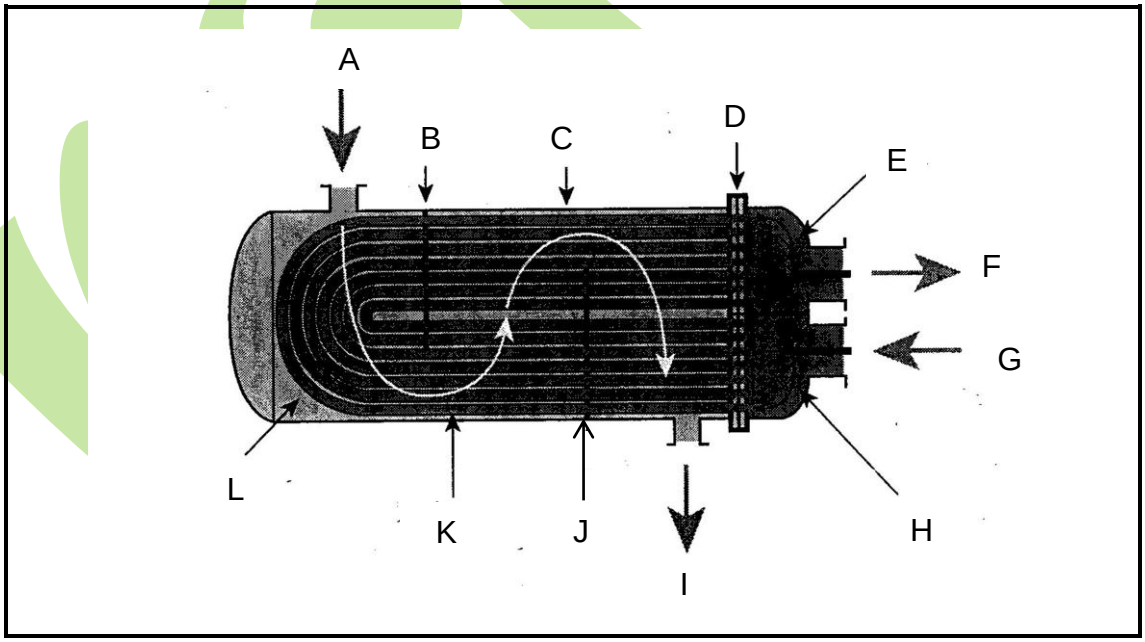
3.2 Which SIX guidelines can be followed when collecting gas samples? (6)

3.3 16 g of calcium carbonate ($CaCO_3$) is dissolved to make 75 dm^3 of a solution. The molar mass of $CaCO_3$ is 100 g/mol.

Calculate the concentration of the solution. (5)
[20]

QUESTION 4

4.1 The diagram below shows a U-tube heat exchanger.



Label the U-tube heat exchanger by writing only the answer next to the letter (A–L) in the ANSWER BOOK. (12 × 1) (12)

4.2 What is the main advantage of air coolers/heaters? (2)

4.3 Describe the function of each of the following components of air coolers/heaters:

4.3.1 Fin tubes

4.3.2 Belt drives

(2 × 2) (4)

4.4 Discuss the basic operating principle of a cooling tower. (9)

4.5 Name THREE gases that air is made up of. (3)

4.6 Write a balanced reaction when methane reacts with oxygen. (4)

[34]

QUESTION 5

5.1 Describe the function of each of the following processes:

5.1.1 Drying

5.1.2 Ion exchange

5.1.3 Adsorption

5.1.4 Membrane separation

(4 × 2) (8)

5.2 What is a humidistat used for? (2)

5.3 Describe the working principle or operation of a spray dryer. (8)

5.4 Clotting is one of the factors indicating poor performance of dryers.

5.4.1 Define *clotting*. (2)

5.4.2 What is the effect of clotting on dryers? (2)

5.4.3 Give TWO reasons for clotting in dryers. (2)

5.4.4 How can clotting in dryers be remedied? (3)

5.5 Describe the flotation-separation process as used in water purification. (9)

[36]

TOTAL: 100

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